

The Level-Six Fibre Lattice and Two-Fibre Linking Forms

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Abstract

For the lattice $\Lambda = H_1(F, \mathbb{Z})$ of a smooth complex two-torus fibre in Engel's fibration, we give the two integral monodromy markings and determine the primitive invariant alternating form ξ , up to sign. The distinguished circle class δ_0 and invariant covector ψ satisfy

$$\iota_{\delta_0} \xi = -6\psi.$$

An explicit unimodular basis puts ξ in the form $J \oplus 6J$; hence its discriminant group is $(\mathbb{Z}/6\mathbb{Z})^2$, with a perfect alternating pairing and a six-dimensional irreducible finite Heisenberg representation. We also compute the complete Smith form of a two-fibre presentation, including the zero-determinant case, and identify its determinant with a denominator-cleared Seifert Euler number. For an explicitly constructed effective rank-one Seifert $\mathbb{C}^*$ -bundle with two multiple fibres, we prove that the primitive integral Leray carrier is indexed by $\text{lcm}(r, s)$ and has coefficient $\Delta/\text{gcd}(r, s)$. The full global Seifert class also has a $\mathbb{Z}/\text{gcd}(r, s)$ component which the coarse transgression does not see. We resolve the integral extension hidden by this differential and compute the symmetric torsion linking form of every effective two-fibre filling. For arbitrary integral local data, the doubled presentation has alternating elementary divisors $(d, |\Delta|/d)$, where $d = \text{gcd}(r, s, m, n)$. These pairings have specified, distinct carriers. Engel's rank-four transgression remains a separate $(3, 4)$ calculation.

1 The fibre calculation

The calculation to be explained is the contraction identity

$$\iota_{\delta_0} \xi = -6\psi. \tag{1.1}$$

Its content is not the coefficient alone: δ_0 , ξ , and ψ must be carried by the same integral geometric lattice, with compatible markings and orientations.

Let $f : Y(u) \rightarrow \mathbb{P}^1$ be the complex two-torus fibration described by Engel, and put

$$U = \mathbb{P}^1 \setminus \{0, 1, \infty\}, \quad F = f^{-1}(t), \quad \Lambda = H_1(F, \mathbb{Z}), \quad V = \Lambda^\vee$$

for $t \in U$. Over U this is also the smooth fibre lattice of the unmodified fibration $X(u) \rightarrow \mathbb{P}^1$. Engel's construction supplies the filtration

$$0 \longrightarrow \Lambda^\times \longrightarrow \Lambda \xrightarrow{\psi} \mathbb{Z} \longrightarrow 0, \quad 0 \longrightarrow \mathbb{Z}\delta_0 \longrightarrow \Lambda^\times \longrightarrow \Lambda' \longrightarrow 0, \tag{1.2}$$

where δ_0 is the circle class of the principal $\mathbb{G}_m$ -bundle in Engel's chosen marking [1, Sections 2 and 5].

We record the integral markings because every sign below depends on them. Let $\varrho = e^{2\pi i/3}$. In the two good-reduction markings, and writing τ_0, τ_1 for the periods of the second elliptic factors, Engel takes

$$\begin{aligned} e_1 &= (1, 0), & e_2 &= (\varrho, 0), & \delta_0 &= (0, 1), & e_4 &= (0, \tau_0), & e_3 &= \tfrac{1}{3}(2 + \varrho, 1), \\ f_1 &= (1, 0), & f_2 &= (i, 0), & \delta_1 &= (0, 1), & f_4 &= (0, \tau_1), & f_3 &= \tfrac{1}{2}(1 + i, 1). \end{aligned} \quad (1.3)$$

The first four entries in each row generate a direct-sum sublattice; after adjoining e_3 , respectively f_3 , the resulting overlattice Λ has ordered bases

$$B_0 = (e_1, e_2, e_3, e_4), \quad B_1 = (f_1, f_2, f_3, f_4).$$

For standard counterclockwise meridians $\gamma_0 \gamma_1 \gamma_\infty = 1$, the matrices act on column coordinates and are

$$[T_0]_{B_0} = \begin{pmatrix} 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad [T_1]_{B_1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.4)$$

The complex orientations give multiplication by ϱ at 0 and by $-i$ at 1. The circle classes and the transition matrix are

$$\delta_0 = -2e_1 - e_2 + 3e_3, \quad \delta_1 = -f_1 - f_2 + 2f_3, \quad C = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (1.5)$$

where the columns of C express B_1 in B_0 and $C\delta_1 = \delta_0$. These are precisely the integral data of Engel's Theorem 5.1. Its proof fixes the sign at infinity by taking a simple zero, rather than a pole, of the linearization for a counterclockwise meridian [1, pp. 12–15].

In the common basis B_0 , direct conjugation and $T_0 T_1 T_\infty = I$ give

$$[T_1]_{B_0} = C[T_1]_{B_1} C^{-1} = \begin{pmatrix} -1 & 1 & -1 & 2 \\ -1 & 0 & -1 & 1 \\ 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (1.6)$$

$$[T_\infty]_{B_0} = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.7)$$

Thus $T_0^3 = T_1^4 = I$, $(T_\infty - I)^2 = 0$, and

$$(T_\infty - I)\Lambda = \langle e_1 - e_3, e_3 - e_2 \rangle. \quad (1.8)$$

Theorem 1.1 (The geometric level-six lattice). *Let $\Gamma = \langle T_0, T_1 \rangle$. For the fibre lattice and markings above,*

$$\Lambda^\Gamma = \mathbb{Z}\delta_0, \quad V^\Gamma = \mathbb{Z}\psi, \quad (\wedge^2 V)^\Gamma = \mathbb{Z}\xi, \quad (1.9)$$

where $\psi = e_4^\vee$ and

$$\xi = 3e_1^\vee \wedge e_2^\vee + e_1^\vee \wedge e_3^\vee - 2e_2^\vee \wedge e_3^\vee - 2e_3^\vee \wedge e_4^\vee. \quad (1.10)$$

With the contraction convention $(\iota_v \xi)(w) = \xi(v, w)$, identity (1.1) holds. Moreover, the ordered basis

$$a_1 = e_1, \quad b_1 = e_3, \quad a_2 = -\delta_0 = 2e_1 + e_2 - 3e_3, \quad b_2 = e_4 - 2e_1 \quad (1.11)$$

is unimodular and satisfies

$$[\xi]_{(a_1, b_1, a_2, b_2)} = J \oplus 6J, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \delta_0 = -a_2, \quad \psi = b_2^\vee. \quad (1.12)$$

Consequently

$$\det(\xi^b) = 36, \quad \text{SNF}(\xi^b) = \text{diag}(1, 1, 6, 6), \quad (1.13)$$

and the elementary divisors of ξ are $(1, 6)$.

Proof. Solving $T_i v = v$ for $i = 0, 1$ gives $v = q(-2, -1, 3, 0)^T$, and solving $T_i^{-T} u = u$ gives $u = q(0, 0, 0, 1)^T$. Integrality makes the two common invariant lattices exactly $\mathbb{Z}\delta_0$ and $\mathbb{Z}\psi$.

Write the coefficients of an alternating form in the order

$$e_1^\vee \wedge e_2^\vee, e_1^\vee \wedge e_3^\vee, e_1^\vee \wedge e_4^\vee, e_2^\vee \wedge e_3^\vee, e_2^\vee \wedge e_4^\vee, e_3^\vee \wedge e_4^\vee.$$

Row reduction of $T_i^T M T_i = M$ gives the integral solutions $q(3, 1, 0, -2, 0, -2)$, proving the last equality in (1.9) and recovering Engel's invariant class from Lemma 7.2 [1, p. 17]. Its matrix in B_0 is

$$[\xi]_{B_0} = \begin{pmatrix} 0 & 3 & 1 & 0 \\ -3 & 0 & -2 & 0 \\ -1 & 2 & 0 & -2 \\ 0 & 0 & 2 & 0 \end{pmatrix}. \quad (1.14)$$

Therefore

$$\delta_0^T [\xi]_{B_0} = (0, 0, 0, -6),$$

which is (1.1).

The columns of (1.11) form the matrix

$$P = \begin{pmatrix} 1 & 0 & 2 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \det P = -1.$$

Multiplication gives $P^T [\xi]_{B_0} P = J \oplus 6J$. This proves the normal form and the Smith form. Finally,

$$\text{Pf}([\xi]_{B_0}) = -6, \quad \frac{1}{2} \xi \wedge \xi = -6e_1^\vee \wedge e_2^\vee \wedge e_3^\vee \wedge e_4^\vee,$$

so $\det[\xi]_{B_0} = (-6)^2 = 36$, as asserted. $\square$

Remark 1.2 (Basis and orientation). If the columns of $A \in \text{GL}_4(\mathbb{Z})$ are a new basis in an old one, then the simultaneous coordinate changes are

$$[\xi]' = A^T [\xi] A, \quad [\delta]' = A^{-1} [\delta], \quad [\psi]' = A^T [\psi]. \quad (1.15)$$

Thus (1.1) is tensorial. The Pfaffian changes by $\det A$, which explains the signs -6 in B_0 and $+6$ in the orientation-reversing basis (1.11), while the determinant and Smith form do not change. Reversing only δ_0 , only ψ , or the contraction convention reverses the sign in (1.1). The abstract normal form alone does not select these classes: for example, in $J \oplus 6J$, taking $\delta = a_1$ and $\psi = b_2^\vee$ gives $\iota_\delta \xi = b_1^\vee$, not -6ψ .

Proposition 1.3 (Reduction along the invariant circle). *Let $K = \ker \psi \subset \Lambda$. The radical of $\xi|_K$ is $\mathbb{Z}\delta_0$, and the induced alternating form on $K/\mathbb{Z}\delta_0$ is unimodular. The classes $[e_1], [e_3]$ form an ordered basis with pairing $\xi([e_1], [e_3]) = 1$.*

Proof. The contraction identity gives $K = \delta_0^\perp$. In the basis (1.11), $K = \mathbb{Z}a_1 \oplus \mathbb{Z}b_1 \oplus \mathbb{Z}a_2$, while $\delta_0 = -a_2$. The restriction of (1.12) to K is $J \oplus 0$, proving every assertion. $\square$

Thus the factor six lies in the plane containing the invariant circle and a vector detected by ψ . Reduction by that circle leaves an integral symplectic plane with elementary divisor one.

2 The discriminant pairing and Heisenberg module

Let (L, ξ) be any rank-four alternating lattice with the normal form $J \oplus 6J$ in a basis (a_1, b_1, a_2, b_2) . Define

$$\xi^\flat : L \longrightarrow L^\vee, \quad \xi^\flat(x) = \xi(x, -), \quad \mathcal{D}_\xi = L^\vee / \xi^\flat(L).$$

The rational extension $\phi = \xi^\flat \otimes \mathbb{Q}$ is an isomorphism.

Lemma 2.1 (Discriminant pairing). *For $u, v \in L^\vee$, the formula*

$$b_\xi(\bar{u}, \bar{v}) = u(\phi^{-1}v) \pmod{\mathbb{Z}} \tag{2.1}$$

defines a perfect alternating pairing $b_\xi : \mathcal{D}_\xi \times \mathcal{D}_\xi \rightarrow \mathbb{Q}/\mathbb{Z}$. Moreover, if

$$\alpha = [a_2^\vee], \quad \beta = [b_2^\vee],$$

then

$$\mathcal{D}_\xi = \langle \alpha \rangle \oplus \langle \beta \rangle \cong (\mathbb{Z}/6\mathbb{Z})^2, \quad b_\xi(\alpha, \beta) = \frac{1}{6}, \quad b_\xi(\beta, \alpha) = -\frac{1}{6}. \tag{2.2}$$

Proof. Write $x = \phi^{-1}u$ and $y = \phi^{-1}v$. Replacing u by $u + \xi^\flat(\lambda)$ changes (2.1) by $\xi(\lambda, y) = -v(\lambda) \in \mathbb{Z}$; replacing v by $v + \xi^\flat(\mu)$ changes it by $u(\mu) \in \mathbb{Z}$. The pairing is therefore well defined and alternating. If it annihilates $\bar{u}$ against every $\bar{v}$, then $v(x) = -u(\phi^{-1}v)$ is integral for every $v \in L^\vee$. Hence $x \in L$, so $\bar{u} = 0$. The induced homomorphism to the Pontryagin dual is injective between finite groups of the same order, and is therefore an isomorphism. Formula (2.2) follows directly from the normal form. $\square$

Put $K = \langle \alpha \rangle \cong \mathbb{Z}/6\mathbb{Z}$, let $\zeta_6 = e^{2\pi i/6}$, and let $\chi \in \widehat{K}$ satisfy $\chi(\alpha) = \zeta_6$. The map

$$r\alpha + s\beta \longmapsto (r\alpha, \chi^s)$$

is a polarization $\mathcal{D}_\xi \cong K \oplus \widehat{K}$. Define

$$H_6 = \mu_6 \times K \times \widehat{K}$$

with multiplication

$$(z, u, \eta)(z', u', \eta') = (zz'\eta'(u), u + u', \eta\eta'). \tag{2.3}$$

Theorem 2.2 (Level-six Heisenberg module). *The center of H_6 is the displayed copy of μ_6 . Its commutator pairing is*

$$[(1, u, \eta), (1, u', \eta')] = (\eta'(u)\eta(u')^{-1}, 0, 1) = (e^{2\pi i b_\xi(d, d')}, 0, 1), \quad (2.4)$$

where $d \leftrightarrow (u, \eta)$ and $d' \leftrightarrow (u', \eta')$. On the six-dimensional space $\mathcal{H} = \mathbb{C}[K]$, the formula

$$[\rho(z, u, \eta)h](t) = z\eta(t)h(t + u) \quad (2.5)$$

defines an irreducible representation, and the central μ_6 acts by its faithful tautological character.

Proof. The bicharacter in (2.3) makes the multiplication associative, and direct multiplication gives (2.4). Its nondegeneracy forces a central element to have $u = 0$ and $\eta = 1$. The same calculation verifies (2.5).

For $t \in K$, let δ_t be the delta-function at t . The operator

$$P_t = \frac{1}{6} \sum_{\eta \in \widehat{K}} \eta(t)^{-1} \rho(1, 0, \eta)$$

projects onto $\mathbb{C}\delta_t$. A nonzero invariant subspace therefore contains some δ_t , and translations carry it through every delta-function. These span $\mathcal{H}$, proving irreducibility. $\square$

Applied to Theorem 1.1, this construction belongs to the fibre lattice $H_1(F, \mathbb{Z})$. It is not constructed from the ordinary cohomology of the total six-manifold.

3 Two multiple fibres

The topology of two logarithmic fillings leads to an elementary presentation whose full integral content is visible before any Leray differential is considered. It is useful to include a global section integer from the start.

Theorem 3.1 (Two-fibre Smith form). *Let $r, s > 0$, let $m, n, b_0 \in \mathbb{Z}$, and define*

$$G_{b_0} = \langle c, x_0, x_1 \mid c \text{ central}, x_0 x_1 = c^{b_0}, x_0^r = c^m, x_1^s = c^n \rangle. \quad (3.1)$$

Put

$$\Delta_{b_0} = rn + sm - rsb_0, \quad d = \gcd(r, s, m, n). \quad (3.2)$$

Then

$$G_{b_0} \cong \text{coker} \begin{pmatrix} r & s \\ -m & n - sb_0 \end{pmatrix}. \quad (3.3)$$

If $\Delta_{b_0} \neq 0$, then

$$G_{b_0} \cong \mathbb{Z}/d\mathbb{Z} \oplus \mathbb{Z}/(|\Delta_{b_0}|/d)\mathbb{Z}, \quad |G_{b_0}| = |\Delta_{b_0}|. \quad (3.4)$$

If $\Delta_{b_0} = 0$, then

$$G_{b_0} \cong \mathbb{Z} \oplus \mathbb{Z}/d\mathbb{Z}. \quad (3.5)$$

In either case G_{b_0} is cyclic if and only if $d = 1$.

Proof. The product relation gives $x_1 = x_0^{-1}c^{b_0}$, so the group is abelian. In additive notation its two remaining relation vectors in $\mathbb{Z}x_0 \oplus \mathbb{Z}c$ are

$$rx_0 - mc, \quad sx_0 + (n - sb_0)c,$$

which proves (3.3). The greatest common divisor of the entries is

$$\gcd(r, s, m, n - sb_0) = \gcd(r, s, m, n) = d,$$

and the determinant is Δ_{b_0} . For nonzero determinant, the Smith invariants are d and $|\Delta_{b_0}|/d$. If the determinant vanishes, the matrix still has rank one because $r, s > 0$, so its Smith form is $\text{diag}(d, 0)$. The cyclicity assertions follow. $\square$

Proposition 3.2 (Section gauges and orientations). *For $k_0, k_1 \in \mathbb{Z}$, the generator change*

$$x'_0 = x_0 c^{k_0}, \quad x'_1 = x_1 c^{k_1} \quad (3.6)$$

sends

$$m' = m + rk_0, \quad n' = n + sk_1, \quad b'_0 = b_0 + k_0 + k_1 \quad (3.7)$$

and preserves both d and Δ_{b_0} . In the gauge $b_0 = 0$, the presentation-preserving choice $k_1 = -k_0 = -k$ sends

$$(m, n) \mapsto (m + rk, n - sk). \quad (3.8)$$

Interchanging (r, m, x_0) with (s, n, x_1) preserves Δ_{b_0} . Replacing the oriented fibre generator by $c' = c^{-1}$ sends (m, n, b_0) to $(-m, -n, -b_0)$ and sends Δ_{b_0} to $-\Delta_{b_0}$.

Proof. Substitution in the three defining relations gives (3.7), and

$$r(n + sk_1) + s(m + rk_0) - rs(b_0 + k_0 + k_1) = \Delta_{b_0}.$$

The remaining statements follow by the indicated changes of generators. They also show why only $|\Delta_{b_0}|$ is intrinsic without fixed orientations. $\square$

The hypotheses in Theorem 3.1 cannot be weakened silently. For $b_0 = 0$,

$$(r, s; m, n) = (2, 4; 2, 2) \implies G_0 \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/6\mathbb{Z},$$

although $\Delta_0 = 12$. Also

$$(2, 4; 2, -4) \implies G_0 \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}, \quad \Delta_0 = 0.$$

At the simplest zero-coefficient boundary, $m = n = 0$ gives $G_0 \cong \mathbb{Z} \oplus \mathbb{Z}/\gcd(r, s)\mathbb{Z}$.

Proposition 3.3 (Seifert Euler number). *Let M be an oriented Seifert manifold over S^2 with two normalized effective exceptional fibres, written in Wilkes's convention as*

$$(b_0, S^2; (r, \beta_0), (s, \beta_1)), \quad (3.9)$$

where $r, s \geq 2$, $0 < \beta_0 < r$, $0 < \beta_1 < s$, and $\gcd(r, \beta_0) = \gcd(s, \beta_1) = 1$. With

$$m = -\beta_0, \quad n = -\beta_1, \quad (3.10)$$

one has $\pi_1(M) \cong G_{b_0}$ and

$$e(M) = -\left(b_0 + \frac{\beta_0}{r} + \frac{\beta_1}{s}\right) = -b_0 + \frac{m}{r} + \frac{n}{s}, \quad rse(M) = \Delta_{b_0}. \quad (3.11)$$

In particular $d = 1$, and hence

$$\pi_1(M) \cong \begin{cases} \mathbb{Z}/|\Delta_{b_0}|\mathbb{Z}, & e(M) \neq 0, \\ \mathbb{Z}, & e(M) = 0. \end{cases} \quad (3.12)$$

If $g = \gcd(r, s)$ and $L = \text{lcm}(r, s) = rs/g$, then the smaller denominator clearing is

$$Le(M) = -Lb_0 + \frac{s}{g}m + \frac{r}{g}n = \frac{\Delta_{b_0}}{g} \in \mathbb{Z}. \quad (3.13)$$

Proof. Wilkes's presentation has central regular fibre h , local relations $a_i^{\alpha_i} h^{\beta_i} = 1$, and, over S^2 with two exceptional fibres, the product relation $a_0 a_1 = h^{b_0}$. Setting $c = h$, $x_i = a_i$ gives (3.1) and (3.10). His Euler-number convention gives (3.11) [5, Proposition 4.3, p. 149]. Effectiveness implies $d = 1$, and Theorem 3.1 gives (3.12). Equation (3.13) is immediate. Hatcher's section-change calculation gives the geometric counterpart of (3.6): twists are added at one boundary torus and subtracted at another, leaving the Euler sum fixed [2, Section 2.1, pp. 25–27]. $\square$

Remark 3.4. The rational number $e(M)$ is orientation-sensitive. Reversing the fibre orientation and renormalizing Wilkes's two local pairs sends

$$(b_0; \beta_0, \beta_1) \mapsto (-b_0 - 2; r - \beta_0, s - \beta_1),$$

which negates both $e(M)$ and Δ_{b_0} but preserves the abstract group. With exceptional fibres, $e(M)$ is a rational Seifert Euler number; only in the ordinary circle-bundle case is it itself the integral Euler class on the underlying base.

4 The rank-one Seifert transgression

The primitive coefficient is visible before passing to Engel's torus fibration. If

$$g = \gcd(r, s), \quad \ell = \text{lcm}(r, s) = \frac{rs}{g},$$

then the two natural integral normalizations are

$$d_2^{0,1}(\ell u) = \frac{\Delta_{b_0}}{g} \omega, \quad d_2^{0,1}(rsu) = \Delta_{b_0} \omega. \quad (4.1)$$

The first uses the primitive global section; the second is its g -fold multiple. We now specify the map and conventions for which these equations hold.

Let $r, s \geq 2$, let $0 < \beta_0 < r$ and $0 < \beta_1 < s$, and assume

$$\gcd(r, \beta_0) = \gcd(s, \beta_1) = 1. \quad (4.2)$$

For $b_0 \in \mathbb{Z}$, consider Kollár's effective Seifert $\mathbb{C}^*$ -bundle

$$q : Z = Y \left(\mathcal{O}_{\mathbb{P}^1}(b_0), \frac{\beta_0}{r}[0] + \frac{\beta_1}{s}[1] \right) \longrightarrow \mathbb{P}^1. \quad (4.3)$$

Equivalently, this is the coarse map of the complement of the zero section in the corresponding line bundle on the labelled root orbisphere. The local characters are faithful by (4.2), so r and s are the actual fibre multiplicities. Give the base its complex orientation and put $\omega = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$. Kollár's natural trivialization

$$\tau : R^1 q_* \mathbb{Q} \xrightarrow{\sim} \mathbb{Q}_{\mathbb{P}^1}$$

defines the normalized rational fibre class $u_K = \tau^{-1}(1)$. We use the Wilkes-oriented class $u = -u_K$ and set

$$m = -\beta_0, \quad n = -\beta_1, \quad e = -b_0 + \frac{m}{r} + \frac{n}{s}, \quad \Delta_{b_0} = rn + sm - rsb_0. \quad (4.4)$$

Theorem 4.1 (Coarse two-fibre transgression). *For the map (4.3), one has $R^0 q_* \mathbb{Z} \cong \mathbb{Z}_{\mathbb{P}^1}$ and an exact sequence of constructible sheaves*

$$0 \longrightarrow R^1 q_* \mathbb{Z} \longrightarrow \mathbb{Z}_{\mathbb{P}^1} \longrightarrow (\mathbb{Z}/r\mathbb{Z})_{\{0\}} \oplus (\mathbb{Z}/s\mathbb{Z})_{\{1\}} \longrightarrow 0, \quad (4.5)$$

where the last map sends 1 to $(1, 1)$. Consequently

$$H^0(\mathbb{P}^1, R^1 q_* \mathbb{Z}) = \ell \mathbb{Z} u, \quad H^1(\mathbb{P}^1, R^1 q_* \mathbb{Z}) \cong \mathbb{Z}/g\mathbb{Z}. \quad (4.6)$$

In the integral Leray spectral sequence of q ,

$$d_2^{0,1} : H^0(\mathbb{P}^1, R^1 q_* \mathbb{Z}) \longrightarrow H^2(\mathbb{P}^1, R^0 q_* \mathbb{Z}), \quad (4.7)$$

the two equations in (4.1) hold.

Proof. Kollár's multiplicity criterion gives the two multiplicities r and s . His integral comparison map and its local cokernel give (4.5) [3, Theorem 7 and Propositions 43 and 47]. The long exact cohomology sequence of (4.5) begins with the diagonal reduction map

$$\mathbb{Z} \longrightarrow \mathbb{Z}/r\mathbb{Z} \oplus \mathbb{Z}/s\mathbb{Z}.$$

Its kernel is $\ell \mathbb{Z}$ and its cokernel is $\mathbb{Z}/g\mathbb{Z}$, proving (4.6).

In Kollár's convention the rational Chern class is

$$c_1^K(Z/\mathbb{P}^1) = \left(b_0 + \frac{\beta_0}{r} + \frac{\beta_1}{s} \right) \omega = -e\omega. \quad (4.8)$$

The quotient $Z \rightarrow Z/\mu_\ell$ induces a map of Leray spectral sequences, and Z/μ_ℓ is an ordinary $\mathbb{C}^*$ -bundle whose Chern class is ℓc_1^K . Hence

$$d_2^{0,1}(\ell u_K) = \ell c_1^K(Z/\mathbb{P}^1)$$

[3, Definition 9, Map 13, and Corollary 44]. Replacing u_K by $u = -u_K$ and using

$$\ell e = -\ell b_0 + \frac{s}{g} m + \frac{r}{g} n = \frac{\Delta_{b_0}}{g}$$

proves the first equation in (4.1). Since $rs = g\ell$, the second follows. $\square$

The coarse coefficient records only the free part of the global Seifert class. To see the missing information integrally, let

$$\mathcal{S}_{r,s} = \langle H, X_0, X_1 \mid rX_0 = H, sX_1 = H \rangle, \quad (4.9)$$

the line-bundle class group of the labelled root orbisphere. Here H is the pullback of the positive coarse class and X_0, X_1 are the two root classes. In the orientation of (4.4), the Seifert class is

$$\varepsilon = -b_0H + mX_0 + nX_1. \quad (4.10)$$

Proposition 4.2 (The global class and its coarse shadow). *There is an isomorphism*

$$\mathcal{S}_{r,s} \cong \mathbb{Z} \oplus \mathbb{Z}/g\mathbb{Z}. \quad (4.11)$$

The section gauges (3.7) leave ε fixed, and

$$\ell\varepsilon = \frac{\Delta_{b_0}}{g}H. \quad (4.12)$$

Equivalently, on gauge classes

$$\kappa[b_0, m, n] = -\ell b_0 + \frac{\ell}{r}m + \frac{\ell}{s}n = \frac{\Delta_{b_0}}{g} \quad (4.13)$$

is a surjection to $\mathbb{Z}$ whose kernel is the torsion subgroup $\mathbb{Z}/g\mathbb{Z}$. Thus the rational Euler number, and hence the coarse differential, determines the full global class exactly when $g = 1$.

Proof. Kollár's classification gives the representatives (b_0, m, n) , while a change of local section gives the relations

$$(b_0, m, n) \sim (b_0 + k_0 + k_1, m + rk_0, n + sk_1).$$

The first determinantal divisor of the relation matrix is 1, while the second is

$$\gcd(r, s, rs) = g.$$

Its Smith form therefore gives (4.11). Relations $rX_0 = H$ and $sX_1 = H$ prove both gauge invariance and (4.12). The three coefficients in (4.13) have greatest common divisor 1, so κ is onto. Its finite kernel has order g , proving the last assertion. $\square$

The torsion is already present in the smallest rank-drop example. For

$$r = s = 2, \quad b_0 = -1, \quad \beta_0 = \beta_1 = 1, \quad (4.14)$$

one has $e = \Delta_{b_0} = 0$, so the coarse transgression vanishes, whereas

$$\varepsilon = H - X_0 - X_1 = X_0 - X_1$$

is the nonzero element of order two in $\mathcal{S}_{2,2}$. Thus a zero rational Euler number need not mean that the global Seifert class is zero.

Proposition 4.3 (Effectivization and rank drop). *For formal local data (r, m) and (s, n) , put*

$$a_0 = \gcd(r, m), \quad a_1 = \gcd(s, n), \quad R = \frac{r}{a_0}, \quad S = \frac{s}{a_1}, \quad h = \gcd(R, S). \quad (4.15)$$

If the data arise from a rank-one Seifert construction after reducing the two local fractions, then the actual local orders are R, S , the primitive carrier is

$$Q = \text{lcm}(R, S) = \frac{rs}{a_0 a_1 h}, \quad (4.16)$$

and

$$d_2^{0,1}(Qu) = \frac{\Delta_{b_0}}{a_0 a_1 h} \omega, \quad d_2^{0,1}(rsu) = \Delta_{b_0} \omega. \quad (4.17)$$

If the local pairs are effective and $\Delta_{b_0} = 0$, then

$$r = s, \quad m + n = r b_0. \quad (4.18)$$

For normalized pairs this says $b_0 = -1$ and $\beta_0 + \beta_1 = r$; in this case $G_{b_0} \cong \mathbb{Z}$.

Proof. The least positive integer k for which $r \mid km$ and $s \mid kn$ is $\text{lcm}(R, S)$, proving (4.16). Writing $m = a_0 M$ and $n = a_1 N$ gives

$$Qe = \frac{\Delta_{b_0}}{a_0 a_1 h}, \quad rs = (a_0 a_1 h)Q,$$

which proves (4.17). If the pairs are effective and $\Delta_{b_0} = 0$, then reduction modulo r gives $r \mid s$, and reduction modulo s gives $s \mid r$. This proves (4.18). The normalized conclusion follows from $0 < \beta_i < r$, and Theorem 3.1 gives the last assertion. $\square$

Remark 4.4 (Naturality, orientations, and scope). The naturality in Theorem 4.1 is the map of Leray spectral sequences induced by $Z \rightarrow Z/\mu_\ell$; the pullback of the ordinary fibre generator is ℓu_K . More generally, a morphism of oriented Seifert bundles gives commuting Leray differentials. If a base pullback has actual carrier $\ell' u'$ with $\ell' \mid \ell$, then the pullback of ℓu is $(\ell/\ell')\ell' u'$, so primitive numerical coefficients must be compared with this index. Reversing the fibre generator or the chosen generator ω reverses the displayed scalar. Section gauges and interchange of the two labelled points preserve (4.1); the only changes of the oriented rank-one fibre basis are $u \mapsto \pm u$.

This theorem is specific to the rank-one Seifert bundle (4.3). An affine torus fibration has a higher-rank local system, and local multiplicities and a group presentation do not determine its extension across singular fibres or its rational transgression. Transporting (4.1) to such a fibration would require an actual map over the base inducing the claimed morphism of constructible sheaves. No such map is used below.

5 The Leray carrier in Engel's (3,4) case

Choose lifts of Engel's logarithmic-transform classes

$$a_0 \in \frac{1}{3}\Lambda^{T_0}, \quad a_1 \in \frac{1}{4}\Lambda^{T_1},$$

and put

$$v_0 = 3a_0, \quad v_1 = 4a_1, \quad m = \psi(v_0), \quad n = \psi(v_1), \quad p = 12\psi(a_0 + a_1) = 4m + 3n. \quad (5.1)$$

We require the local cyclic affine actions on the good-reduction tori to be free. Their quotient fibres then have multiplicities 3 and 4. The condition $|p| = 1$ used below implies this freeness. Exact torsion order alone does not imply it.

Proposition 5.1 (A free-action criterion). *Let $d > 0$ be an integer, let T preserve a lattice Λ with $T^d = I$, and let $a \in d^{-1}\Lambda^T$. On the real torus $\Lambda_{\mathbb{R}}/\Lambda$, put $g_a(z) = Tz + a$. If $\theta \in \Lambda^\vee$ satisfies $\theta T = \theta$ and $\gcd(\theta(da), d) = 1$, then the action generated by g_a is free and has order d . In the two markings above, both local actions are free if and only if $\gcd(m, 3) = \gcd(n, 4) = 1$, equivalently $\gcd(p, 12) = 1$.*

Proof. Since $Ta = a$, one has $g_a^k(z) = T^k z + ka$ and $g_a^d = 1$. A fixed point of g_a^k would imply $(T^k - I)z + ka \in \Lambda$. Applying the real extension of θ would give $k\theta(da)/d \in \mathbb{Z}$. The coprimality assumption therefore forces $d \mid k$.

In the first marking, direct solution of $T_0 v = v$ gives $3a_0 = h\delta_0 + me_4$ with $h \in \mathbb{Z}$. If $3 \mid m$, then $z = -h(e_1 + e_2)/3$ satisfies $(T_0 - I)z + a_0 = he_3 + (m/3)e_4 \in \Lambda$. In the second marking, similarly $4a_1 = h'\delta_1 + nf_4$. If $2 \mid n$, then $z = -h'(f_1 + f_2)/4$ satisfies $(T_1^2 - I)z + 2a_1 = h'f_3 + (n/2)f_4 \in \Lambda$. These fixed points prove necessity. Sufficiency follows from the first assertion with $\theta = \psi$. The final equivalence follows from $p \equiv m \pmod{3}$ and $p \equiv -n \pmod{4}$. $\square$

The class $a_0 = \delta_0/3$ has exact order three, but its affine action fixes $z = -(e_1 + e_2)/3$, since

$$(T_0 - I)z + a_0 = e_3 \in \Lambda.$$

Likewise $a_1 = \delta_1/4$ has exact order four, but its square fixes $z = -(f_1 + f_2)/4$, because

$$(T_1^2 - I)z + 2a_1 = f_3 \in \Lambda.$$

These computations specify the needed free-action hypothesis in the local quotient construction [1, Section 3.2, pp. 10–11].

The carrier of the transgression is the cohomological Leray spectral sequence of the specific map $f : Y(u) \rightarrow \mathbb{P}^1$,

$$E_2^{a,b} = H^a(\mathbb{P}^1, R^b f_* \mathbb{Z}) \implies H^{a+b}(Y(u), \mathbb{Z}). \quad (5.2)$$

Over U , the restriction of $R^1 f_* \mathbb{Z}$ is a local system with fibre $V = H^1(F, \mathbb{Z})$ and monodromy T_i^{-T} . Across the three singular values, $R^1 f_* \mathbb{Z}$ is the corresponding constructible sheaf, not a local system. The fibres of f are connected, so $R^0 f_* \mathbb{Z} \cong \mathbb{Z}_{\mathbb{P}^1}$. Thus the first relevant differential is precisely

$$d_2^{0,1} : H^0(\mathbb{P}^1, R^1 f_* \mathbb{Z}) \longrightarrow H^2(\mathbb{P}^1, R^0 f_* \mathbb{Z}). \quad (5.3)$$

Theorem 5.2 (Engel's integral transgression). *Assume $|p| = 1$, as in Engel's integral-homology calculation, and choose the common orientation for which $H^2(\mathbb{P}^1, R^0 f_* \mathbb{Z}) = \mathbb{Z}\omega$. Then*

$$H^0(\mathbb{P}^1, R^1 f_* \mathbb{Z}) = 12\mathbb{Z}\psi, \quad d_2^{0,1}(12\psi) = p\omega. \quad (5.4)$$

Proof. The extension lattice in (5.4) is Engel's Lemma 7.5. For the differential, the two local affine lifts satisfy

$$\tilde{\rho}_0^3 = t_{v_0}, \quad \tilde{\rho}_1^4 = t_{v_1}.$$

Their oriented contributions to the obstruction for 12ψ are

$$\frac{12\psi(v_0)}{3} = 4m, \quad \frac{12\psi(v_1)}{4} = 3n. \quad (5.5)$$

The cusp gluing at infinity contributes zero in this marking. Their sum is $p = 4m + 3n$, which is Engel's Lemma 7.6 [1, pp. 20–21]. Reversing either generator in (5.4) reverses the corresponding sign; simultaneous transport of the orientation preserves the identity. $\square$

Proposition 5.3 (The (3, 4) presentation). *For the same geometric data,*

$$\pi_1(Y(u)) \cong \langle c, x_0, x_1 \mid c \text{ central}, x_0x_1 = 1, x_0^3 = c^m, x_1^4 = c^n \rangle. \quad (5.6)$$

Consequently

$$\pi_1(Y(u)) \cong \begin{cases} \mathbb{Z}/|p|\mathbb{Z}, & p \neq 0, \\ \mathbb{Z}, & p = 0. \end{cases} \quad (5.7)$$

In particular it is trivial when $|p| = 1$.

Proof. Engel's vanishing cycles at infinity kill $\ker \psi$ and leave the central class c represented by e_4 . The two logarithmic fillings give (5.6) [1, Proposition 6.1, pp. 15–16]. This is Theorem 3.1 with $(r, s; b_0) = (3, 4; 0)$, for which $d = 1$ and $\Delta_0 = p$. $\square$

Comparison with the rank-one theorem

For $(r, s; b_0) = (3, 4; 0)$, one has $g = 1$, $\ell = 12$, and $\Delta_0 = 4m + 3n = p$. Thus Theorem 4.1 and (5.4) have the same scalar after the rank-one class u and Engel's invariant covector ψ are given the same sign. The equality is only numerical. Engel's sheaf has generic fibre $H^1(F, \mathbb{Z})$ of rank four, nontrivial monodromy, and an additional singular value at infinity. No factorization of f through (4.3), nor a morphism carrying u to ψ across all singular fibres, has been constructed here. Engel's Lemmas 7.5–7.6, rather than Theorem 4.1, prove the carrier and differential in (5.4).

Likewise, the presentation (3.1) alone defines neither a map, the sheaves $R^b f_* \mathbb{Z}$, nor a rational transgression. It cannot imply a theorem for an arbitrary affine torus fibration. Such a result requires the full local system, every specialization map, a global affine or Chern class, and compatible orientations.

Finally, $\Lambda = H_1(F, \mathbb{Z})$ is an auxiliary fibre lattice. In the case in which Engel's Theorem 4.1 identifies the total space with S^6 , one has

$$H^j(Y(u), \mathbb{Z}) = 0 \quad (0 < j < 6),$$

so neither Λ nor its discriminant group is ordinary positive-degree cohomology of the total space.

6 The integral extension of the two-fibre class

The coefficient Δ_{b_0}/g determines the coarse transgression. The total integral group can still be a nonsplit extension. For example, $r = s = 2$, $b_0 = 0$, and $m = n = -1$ give a cyclic group of order four, although both successive quotients have order two.

Retain the group $\mathcal{S}_{r,s}$ in (4.9), for arbitrary $r, s > 0$ and $m, n, b_0 \in \mathbb{Z}$. Write

$$g = \gcd(r, s), \quad R = r/g, \quad S = s/g, \quad \ell = gRS, \quad k = \Delta_{b_0}/g.$$

Choose integers a, b with $aS + bR = 1$, and put

$$U = aX_0 + bX_1, \quad T = RX_0 - SX_1, \quad t = bm - an. \quad (6.1)$$

The letters a, b in this section denote Bézout coefficients.

Theorem 6.1 (Integral extension and splitting). *One has*

$$\mathcal{S}_{r,s} = \mathbb{Z}U \oplus (\mathbb{Z}/g\mathbb{Z})T, \quad H = \ell U, \quad \varepsilon = kU + tT. \quad (6.2)$$

Consequently, for $E = \mathcal{S}_{r,s}/\mathbb{Z}\varepsilon$,

$$E = \langle U, T \mid gT = 0, kU + tT = 0 \rangle. \quad (6.3)$$

If $k \neq 0$, this gives an exact sequence

$$0 \longrightarrow \mathbb{Z}/g\mathbb{Z} \xrightarrow{1 \mapsto T} E \xrightarrow{U \mapsto 1, T \mapsto 0} \mathbb{Z}/|k|\mathbb{Z} \longrightarrow 0. \quad (6.4)$$

It splits if and only if $\gcd(g, k)$ divides t . Moreover, with $d = \gcd(r, s, m, n)$,

$$\gcd(g, k, t) = d, \quad E \cong \begin{cases} \mathbb{Z}/d\mathbb{Z} \oplus \mathbb{Z}/(|\Delta_{b_0}|/d)\mathbb{Z}, & k \neq 0, \\ \mathbb{Z} \oplus \mathbb{Z}/d\mathbb{Z}, & k = 0. \end{cases}$$

Thus E and G_{b_0} have the same invariant factors. This assertion alone specifies no canonical isomorphism between their generators.

Proof. Eliminate H from (4.9). The remaining relation is $g(RX_0 - SX_1) = 0$. The change of generators in (6.1) has determinant $-aS - bR = -1$, with inverse

$$X_0 = SU + bT, \quad X_1 = RU - aT.$$

It follows that T has exact order g and $H = rX_0 = \ell U$. Substitution in (4.10) gives (6.2) and (6.3).

If $k \neq 0$, no nonzero multiple of the relation $kU + tT$ lies in $\mathbb{Z}T$. Thus the indicated cyclic subgroup injects, and the quotient is $\mathbb{Z}/|k|\mathbb{Z}$. A lift of its generator has form $U + jT$ and is killed by k precisely when $kj \equiv t \pmod{g}$. This congruence is soluble exactly under the stated divisibility condition.

The matrix sending (m, n) to $(Sm + Rn, bm - an)$ is unimodular. Since $g \mid \ell$, this gives

$$\gcd(g, k, t) = \gcd(g, Sm + Rn, bm - an) = \gcd(g, m, n) = d.$$

The relation matrix in (6.3) has nonzero entry g , determinant of absolute value $g|k| = |\Delta_{b_0}|$, and first determinantal divisor d . Its Smith form proves both cases. $\square$

Replacing (a, b) by $(a + Rj, b - Sj)$ changes U to $U + jT$ and changes t to $t - jk$ modulo g . The presentation and the splitting condition are therefore independent of the Bézout choice. The section gauges (3.7) change t by a multiple of g and leave k fixed.

Proposition 6.2 (The coarse Leray extension). *For the effective rank-one Seifert bundle (4.3), let M be its oriented unit-circle subbundle. If $k \neq 0$, the integral Leray filtration gives*

$$0 \longrightarrow \mathbb{Z}/|k|\mathbb{Z} \longrightarrow H^2(M, \mathbb{Z}) \longrightarrow \mathbb{Z}/g\mathbb{Z} \longrightarrow 0. \quad (6.5)$$

Its middle group is cyclic of order $|\Delta_{b_0}|$. The extension splits if and only if $\gcd(g, k) = 1$. For $k = 0$, the middle group is $\mathbb{Z}$ and the sequence is $0 \rightarrow \mathbb{Z} \xrightarrow{\pm g} \mathbb{Z} \rightarrow \mathbb{Z}/g \rightarrow 0$.

Proof. A neighbourhood of each fibre retracts onto a circle, so $R^j q_* \mathbb{Z} = 0$ for $j > 1$. The base has cohomological dimension two. Hence the only terms contributing to total degree two are $E_{\infty}^{2,0} = \text{coker } d_2^{0,1}$ and $E_{\infty}^{1,1} = \mathbb{Z}/g\mathbb{Z}$. Equation (4.1) gives the left group in (6.5). Radial contraction from Z to M preserves the base and its Leray filtration.

Effectiveness implies $d = 1$. Theorem 3.1 and abelianity give $H_1(M, \mathbb{Z}) = G_{b_0}$, and integral Poincaré duality gives $H^2(M, \mathbb{Z}) \cong H_1(M, \mathbb{Z})$. If $k \neq 0$, a direct sum of cyclic groups of orders g and $|k|$ is cyclic exactly when those orders are coprime. Conversely, a cyclic group of order $g|k|$ has complementary subgroups of these orders when they are coprime. This proves the splitting assertion. If $k = 0$, Theorem 3.1 gives the middle group $\mathbb{Z}$. Its subgroup with quotient $\mathbb{Z}/g\mathbb{Z}$ is $g\mathbb{Z}$. $\square$

The algebraic sequence (6.4) places the two cyclic groups in the opposite order from (6.5). Their middle groups have the same invariant factors. No identification of the two filtrations is asserted. For $r = s = 2$, $m = n = -1$, choose $a = 1, b = 0$. Then $t = 1$ and $k = -2(b_0 + 1)$. At $b_0 = 0$, both extensions have middle group $\mathbb{Z}/4\mathbb{Z}$ and do not split. At $b_0 = -1$, the nonzero torsion class $\varepsilon = T$ has zero coarse coefficient, and quotienting by it leaves $E \cong \mathbb{Z}$.

7 The linking form of an effective filling

A group presentation determines the orders of cycles. Their linking also uses the oriented filling slopes. We retain this geometric data explicitly, without selecting a convention for naming lens spaces.

Let T^2 have oriented basis (x, c) with intersection $x \cdot c = 1$. Set $\nu = n - sb_0$ and regard

$$\mu_0 = rx - mc, \quad \mu_1 = sx + \nu c \quad (7.1)$$

as curves on T^2 . Suppose $\gcd(r, m) = \gcd(s, n) = 1$. Both curves are then primitive. Take an oriented solid torus V_0 whose meridian is μ_0 , identifying T^2 with its boundary with the boundary orientation. Glue a second solid torus V_1 with meridian μ_1 by an orientation-reversing boundary map. Denote the resulting oriented closed manifold by M_A . The slope c specifies its regular Seifert fibre. The intersection numbers with the two meridians have absolute values r, s . The section relation $x_1 = -x + b_0 c$ identifies the second local relation with $-\mu_1$. Thus these fillings realize the data of (3.1), with the stated orientation on T^2 . The gluing description is the two-fibre case of the solid-torus construction in [2, Section 2.1, pp. 25–27].

For disjoint torsion cycles z, w in an oriented closed three-manifold, our convention is

$$\lambda_M([z], [w]) = \frac{C \cdot w}{N} \pmod{\mathbb{Z}}, \quad \partial C = Nz, \quad N \neq 0.$$

This is the symmetric torsion linking pairing [4, Section 1, pp. 2–3].

Theorem 7.1 (Two-fibre linking coefficient). *Choose integers α, β with $m\alpha + r\beta = 1$, and put*

$$\lambda_0 = \alpha x + \beta c, \quad q = s\beta - \nu\alpha, \quad \Delta = rn + sm - rsb_0. \quad (7.2)$$

Then

$$\mu_1 = q\mu_0 + \Delta\lambda_0, \quad \gcd(q, \Delta) = 1.$$

If $\Delta \neq 0$, the class $\gamma = [\lambda_0]$ generates $H_1(M_A, \mathbb{Z}) \cong \mathbb{Z}/|\Delta|\mathbb{Z}$, and

$$\lambda_{M_A}(i\gamma, j\gamma) = -\frac{qij}{\Delta} \pmod{\mathbb{Z}}, \quad [x] = m\gamma, \quad [c] = r\gamma. \quad (7.3)$$

The pairing is nonsingular. For $\Delta = 0$, one has $H_1(M_A, \mathbb{Z}) \cong \mathbb{Z}$ and the torsion pairing is trivial.

Proof. The columns (μ_0, λ_0) form a unimodular matrix of determinant $r\beta + m\alpha = 1$. Taking determinants in this basis gives

$$\mu_1 = (s\beta - \nu\alpha)\mu_0 + (r\nu + sm)\lambda_0.$$

Primitivity of μ_1 proves $\gcd(q, \Delta) = 1$. The inverse change of basis is

$$x = \beta\mu_0 + m\lambda_0, \quad c = -\alpha\mu_0 + r\lambda_0.$$

The two meridional disks kill μ_0 and μ_1 in first homology, so the resulting presentation has the single relation $\Delta\gamma = 0$.

Let D_0, D_1 be the oriented meridional disks with respective boundaries μ_0, μ_1 . On T^2 , the cycle $\mu_1 - q\mu_0 - \Delta\lambda_0$ bounds an integral singular 2-chain B , because its homology class is zero. Consequently

$$C = D_1 - qD_0 - B, \quad \partial C = \Delta\lambda_0.$$

Push a representative of λ_0 into the interior of V_0 . It is disjoint from D_1 and B and intersects D_0 with sign $+1$. The sign follows from the boundary orientation and $\mu_0 \cdot \lambda_0 = 1$. Therefore $C \cdot \lambda_0 = -q$, which proves (7.3). Bilinearity gives all values. Since q is a unit modulo Δ , this pairing identifies the cyclic group with its character group. If $\Delta = 0$, its first homology is infinite cyclic and has no torsion. $\square$

Replacing (α, β) by $(\alpha + rj, \beta - mj)$ replaces λ_0 by $\lambda_0 + j\mu_0$ and q by $q - j\Delta$. Thus neither γ nor the linking value changes. Reversing the orientation of M_A negates the pairing. A different cyclic generator multiplies its displayed coefficient by a unit square modulo $|\Delta|$. These rules state the full dependence on choices, including even orders.

For $(r, s; m, n; b_0) = (2, 2; -1, -1; 0)$, take $(\alpha, \beta) = (-1, 0)$. Then $\Delta = -4$, $q = -1$, and $\lambda_{M_A}(\gamma, \gamma) = -1/4$. For $(r, s; m, n; b_0) = (2, 3; 1, 1; 0)$, take $(\alpha, \beta) = (1, 0)$. Now $\Delta = 5$, $q = -1$, and the value is $1/5$. The nonsingular symmetric form on $\mathbb{Z}/5\mathbb{Z}$ cannot be an alternating form: every alternating bilinear form on a cyclic group vanishes. The next construction specifies the larger group on which an alternating form does exist.

8 Alternating doubles of the arithmetic presentation

Return to arbitrary $r, s > 0$ and $m, n, b_0 \in \mathbb{Z}$, without local coprimality assumptions. Regard the actual relation matrix

$$A = \begin{pmatrix} r & s \\ -m & n - sb_0 \end{pmatrix} : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$$

as an integral map. On $L_A = \mathbb{Z}^2 \oplus \mathbb{Z}^2$, define

$$\Omega_A((u, v), (u', v')) = u^T A v' - (u')^T A v, \quad [\Omega_A] = \begin{pmatrix} 0 & A \\ -A^T & 0 \end{pmatrix}. \quad (8.1)$$

This construction uses the full presentation matrix and both copies of $\mathbb{Z}^2$ as specified. It is an alternating lattice associated with the presentation, independently of a geometric fibre lattice.

Theorem 8.1 (Arithmetic discriminant and its pairing). *Put $d = \gcd(r, s, m, n)$ and $\Delta = \det A$. If $\Delta \neq 0$, set $N = |\Delta|/d$. Then*

$$(L_A, \Omega_A) \cong (\mathbb{Z}^4, dJ \oplus NJ), \quad \mathcal{D}_A \cong (\mathbb{Z}/d\mathbb{Z})^2 \oplus (\mathbb{Z}/N\mathbb{Z})^2.$$

Here $\mathcal{D}_A = L_A^\vee / \Omega_A^\flat(L_A)$ and $\Omega_A^\flat(z) = \Omega_A(z, -)$. In the coordinates

$$\mathcal{D}_A = \text{coker } A \oplus \text{coker } A^T,$$

its perfect alternating pairing is

$$b_A((\bar{x}, \bar{y}), (\bar{x}', \bar{y}')) = x^T A^{-T} y' - (x')^T A^{-T} y \pmod{\mathbb{Z}}. \quad (8.2)$$

Its exponent is N and its order is Δ^2 . If $\Delta = 0$, the radical of Ω_A is primitive of rank two, and its quotient lattice has form dJ and discriminant group $(\mathbb{Z}/d\mathbb{Z})^2$.

Proof. Choose unimodular matrices P, Q with $PAQ = \text{diag}(d, N)$ when $\Delta \neq 0$. The change of basis $\text{diag}(P^T, Q)$ in (8.1), followed by interleaving the two factors, gives $dJ \oplus NJ$. The integer d divides N because d^2 divides $\det A$. In the original coordinates the matrix of $\Omega_A^\flat$ is $-[\Omega_A]$. Its inverse over $\mathbb{Q}$ is

$$\begin{pmatrix} 0 & A^{-T} \\ -A^{-1} & 0 \end{pmatrix},$$

which gives (8.2) by (2.1). More explicitly, $x^T A^{-T} y$ is integral on either relation lattice: replace x by $x + Az$ or y by $y + A^T w$. If it is integral for every integral y , then $A^{-1}x$ is integral, so $x \in A\mathbb{Z}^2$. The same argument applies to y . This proves perfectness directly and determines the stated orders.

If $\Delta = 0$, the matrix A has rank one since $r, s > 0$. Its Smith form is $\text{diag}(d, 0)$, so the same change of basis gives $dJ \oplus 0_2$. The last summand is the radical and is a direct summand of L_A . Taking its quotient proves the last assertion. The unreduced cokernel also has a free summand $\mathbb{Z}^2$. It is not the finite discriminant group of a nondegenerate lattice. $\square$

Let $K = \text{coker } A$ when $\Delta \neq 0$. The character

$$\chi_{\bar{y}}(\bar{x}) = \exp(2\pi i x^T A^{-T} y)$$

identifies $\text{coker } A^T$ with $\widehat{K}$. Thus (8.2) is the standard alternating pairing on $K \oplus \widehat{K}$. Define

$$H_A = \mu_N \times K \times \widehat{K}, \quad (z, u, \eta)(z', u', \eta') = (zz'\eta'(u), u + u', \eta\eta').$$

On $\mathbb{C}[K]$, the formula

$$[\rho(z, u, \eta)h](v) = z\eta(v)h(v + u)$$

gives a representation of dimension $|\Delta|$. Its central character is tautological, its commutator is $\exp(2\pi i b_A)$, and its center is μ_N . Indeed, perfectness forces any central element to have $u = 0$ and $\eta = 1$. Character averaging projects onto each delta-function, and translations act transitively on those functions. This proves irreducibility exactly as in Section 2, for every finite abelian K , including groups of even exponent. For $\Delta = 0$, the nondegenerate quotient instead gives $K = \mathbb{Z}/d\mathbb{Z}$ and a representation of dimension d .

For the non-effective data $(2, 4; 2, 2; 0)$, one has $d = 2$, $\Delta = 12$, and alternating elementary divisors $(2, 6)$. Thus $\mathcal{D}_A \cong (\mathbb{Z}/2)^2 \oplus (\mathbb{Z}/6)^2$, and the representation has dimension twelve and central order six. For $(2, 4; 2, -4; 0)$, the radical has rank two and the quotient has form $2J$. The resulting finite group is $(\mathbb{Z}/2)^2$. These are arithmetic constructions. Nonprimitive filling slopes do not bound meridional disks of a solid torus, so Theorem 7.1 does not apply to these unreduced data.

Proposition 8.2 (Separation from the level-six fibre). *For effective two-fibre data with $(r, s) = (3, 4)$ and $b_0 = 0$, the arithmetic double has elementary divisors $(1, |p|)$, and $\gcd(p, 12) = 1$. In particular, it never has elementary divisors $(1, 6)$. When $|p| = 1$, its discriminant group is zero, while the fibre lattice of Theorem 1.1 has discriminant group $(\mathbb{Z}/6)^2$.*

Proof. Here $d = 1$ and $\Delta = p = 4m + 3n$. Effectiveness makes m a unit modulo three and n a unit modulo four. Reduction of p modulo three and four proves $\gcd(p, 12) = 1$. Theorem 8.1 now gives the result. $\square$

The abstract data $(3, 4; 3, -2; 0)$ do have $\Delta = 6$ and $d = 1$. Their doubled lattice is therefore isometric to $J \oplus 6J$, but both local pairs are non-effective. This is an equality of abstract lattice types. It supplies no map to Engel's fibre or to its Leray sheaf.

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